

FIE401: Financial Econometrics

Oral Exam Topics

Week 3: Simple regression

1. OLS.
2. How do we interpret the coefficient of interest?
3. How do we test whether a coefficient is different from zero?
4. Economic significance.
5. How do we measure goodness of fit?

Week 4: Multiple regression

6. Why do we include control variables in regressions?
7. How do we interpret the coefficient of interest in a multiple regression?
8. What is adjusted R-squared? Why do we need it?
9. What is F-test? Why is t-test not enough?
10. Which hypotheses can we test with F-test?

Week 5: Assumptions and biases

11. OLS assumption 1: Zero conditional mean.
12. OLS assumption 2: Random sampling.
13. OLS assumption 3: Rare outliers.
14. OLS assumption 4: No multicollinearity.
15. VIF.
16. Which methods are used to deal with outliers?
17. OLS property 1: Unbiased.
18. OLS property 2: Consistent.
19. OLS property 3: Normally distributed.
20. Heteroscedasticity of errors.
21. Biases: Sample selection bias.
22. Biases: Omitted variable bias.
23. Biases: Simultaneity bias.
24. Biases: Attenuation bias.

Week 6: Non-linear functional form

25. Polynomials. How do we interpret coefficients?
26. Log transformation. How do we interpret coefficients?
27. Interactions. How do we interpret coefficients?

Week 7: Binary dependent variable

28. What is a binary variable? Give an example.

29. Linear probability model (LPM). How do we interpret coefficient of interest?
30. Logit and probit regressions. What are their benefits relative to LPM?
31. How can you estimate marginal effects of logit and probit regressions?
32. What does MLE stand for? What is the intuition behind its construction?
33. What is McFadden R-squared? Why is simple R-squared bad?
34. How do you test hypotheses in logit and probit?

Week 8: Panel data analysis

35. What is cross-sectional data, time-series data, and panel data?
36. First Difference Estimator.
37. Pooled OLS.
38. Entity-demeaned estimator.
39. How do we interpret coefficient of interest in regressions with entity fixed effects?
40. How can we deal with fixed effects?
41. Zero conditional mean assumption. What changes relative to simple OLS assumption?
42. Clustered standard errors.

Week 9: Event study

43. Price-based event studies vs value event studies.
44. Which steps should you undertake to perform an event study?
45. How does a usual timeline in an event study look like?
46. Why do we need an estimation window?
47. How do we assess whether an event leads to a significant market reaction?

Week 10: Instrumental variables

48. What do abbreviations “IV” and “2SLS” stand for?
49. Why do we need an IV regression? Which issues can it solve?
50. Good instrument: Relevance.
51. How can we check whether an instrument is relevant?
52. Good instrument: Exogeneity.
53. How can we check instruments’ exogeneity?
54. Why should you estimate 2SLS regression in one go?
55. How many instruments do you need?

Week 11: Difference in differences estimator

56. Why are experiments useful?
57. What is the difference between experiments and quasi-experiments?
58. What is internal validity?
59. What is external validity?
60. Which assumption should hold if you use difference in differences estimator?
61. Can you test the parallel trend assumption?
62. What is matching?
63. Name two different matching algorithms